Referee report on `mjl_part4_draft3.tex`

"A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, IV", Draft 3

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0. Scope

Read in full. Draft 3 (1090 lines; 16 pages; it compiles cleanly) and its revision log.

Compiled numbering.

- Proposition 2.1;
- Theorem 3.1, Example 3.2, Remark 3.3;
- Lemma 4.1, Proposition 4.2, Remark 4.3;
- Proposition 5.1, Theorem 5.2;
- Proposition 6.1;
- Proposition 7.1;
- Definition 8.1, Lemma 8.2, Proposition 8.3.

Which versions I could check. Draft 3 now cites Part I, Draft 7, Part II, Draft 7, and Part III, Draft 9. I have only Draft 6 of each.

- Every cited number agrees with Draft 6:
  - Part I: Propositions 2.3, 2.6, 3.1, 3.5, 5.5 and 6.1; Lemma 5.1; Example 3.7; Theorem 6.6;
  - Part II: Definition 2.1, Theorem 6.2;
  - Part III: Lemma 3.3; Theorems 4.3, 7.6 and 9.1; Corollaries 5.1 and 5.2; Lemma 7.5; Proposition 11.3.
- I could not check Part III, Corollary 7.8 (new in Draft 9), or the new subtitles of Parts I and II.
- §8.4 now cites "Part I, Proposition 2.3 and its archimedean remark". I could not identify that remark in Draft 6. The complex-place splitting is (R4) of Part I, Appendix B, according to the revision log. I did not read that appendix.

External sources. The only new external citation is Clozel–Delorme. Its bibliographic data agree with the standard citation as far as I know; I did not check the paper. I have not checked Li's paper, so I cannot confirm that its Paley–Wiener hypothesis is stated only at *real* places.

Summary verdict

Every point of my Draft 2 report has been addressed correctly. I rechecked each repair:

- the new Haar-modulus proof (Proposition 4.2), which is now correct and complete;
- the justification of $u(x) = 1$;
- Remark 3.3 and its field-torus proof;
- the central-character discussion in Proposition 7.1;
- the Li/Clozel–Delorme paragraph;
- the merged spectral remark.

Sections 2–7 are now correct and essentially ready. I have only minor editorial points on them (§3).

§8 is the new part, and it is correct as stated. I found no error in Definition 8.1, Lemma 8.2 or Proposition 8.3. But I have four substantive comments (§1). One affects how Proposition 8.3 should be read: its product condition (iii) is a *normalization*, not an existence problem. Another identifies a case that looks provable now: $r = 2$.

On "getting closer to the goals". Two kinds of progress should be kept apart.

- *As a paper,* Draft 3 is close to finished. Its claims are correct, correctly labeled and correctly attributed.
- *As progress on the series' two open goals,* it has not proved either of them:
  - integrality of L_D at wild places;
  - the comparison of $D^\times$ with its cover.

  What it has done, and this is real progress, is reduce the second goal to a short list of precise local statements at split places. For $r = 2$ part of that list already appears to hold (§1.3).

1. Section 8

1.1 The product condition in Proposition 8.3 is a normalization

Suppose (i) and (ii) hold, with every λ_v independent of the class. Then the proof of Proposition 8.3 gives, with no further hypothesis,

$$I_{\widetilde{H}}(\widetilde{f}) = \Lambda\, I_H(f), \qquad \Lambda = \prod_{v \notin S} \lambda_v \prod_{v \in S} b_{D,v}.$$

Hypothesis (iii), $\Lambda = 1$, can always be arranged by rescaling one test function. So it is not a constraint on existence.

What matters is the *value* of Λ under canonical normalizations: L_D (unitary) at S, and the Flicker–Kazhdan unit elements and matching at $v \notin S$. Integrality of the global multiplicities then constrains the local character coefficients.

I recommend restating the conclusion as $I_{\widetilde{H}} = \Lambda I_H$, with Λ written out, and noting that Λ is computable once the split-place normalization is fixed. The real hypothesis in (i) is *uniformity*: λ_v must not depend on s_v, and in particular not on the torus type k_v.

1.2 The trace formulas are simpler than Proposition 8.3 assumes

Both sides are compact-quotient trace formulas: D is a division algebra, and the cover is compact modulo the central preimage. Since r is prime, every noncentral element of $D^\times(K)$ is regular elliptic. Consequently:

- there are no singular noncentral classes;
- there is no continuous spectrum;
- absolute convergence holds for every smooth compactly supported test function;
- the only term the support condition must remove is the identity term.

The convergence clause in (iii) and the talk of "singular terms" can therefore be dropped.

Better: *keep* the identity term, and use it as an independent check on Λ. The relevant facts:

- the adelic quotients $Z(\mathbb{A})\backslash\overline{G_D}(\mathbb{A})$ modulo $\iota(D^\times(K))$ and $\mathbb{A}^\times D^\times(K)\backslash D^\times(\mathbb{A})$ are homeomorphic, so they have equal volumes;
- at $v \in S$, $\widetilde{f}_v(1) = c_{D,v} f_v(1)$ (Part III, Corollary 9.3).

On the regular terms the constant at $v \in S$ is $b_{D,v}$, while on the identity term it is $c_{D,v} = b_{D,v}|n|_v^{-r(r-1)/2}$. So the identity terms must supply the extra factor $\prod_{v\in S}|n|_v^{-r(r-1)/2}$, balanced by the split places.

The ratio $\Delta(t)/\Delta(t^n)$ tends to $|n|_v^{-r(r-1)/2}$ near scalars for *every* torus type. The proof of Part III, Lemma 3.3, only uses $(1-u^n)/(1-u) \to n$. This suggests that the split-place identity values contribute $|n|_v^{-r(r-1)/2}$ as well. If so, the product formula $\prod_v |n|_v = 1$ balances the identity terms, provided the complex places are included, where $|n|_v = n^2$.

This is heuristic. But it is a concrete consistency test that the draft could record, and it uses the identity-term comparison of Part I, Proposition 6.1.

1.3 Definition 8.1 names the wrong "in particular"

The example given, $J(\xi x, f_v) = J(x, f_v)$ for scalar $\xi \in \mu_n$, is *automatic*. The ordinary test function has central character θ^{-1}, and $\theta(\xi) = \tilde{\theta}(\xi^*) = 1$ (§2.2). So Lemma 8.2's "in particular" case has no content.

The real constraint is $x' = \zeta x$ with $\zeta \in \mu_n(E_v)$ *nonscalar*, i.e. ζ a nontrivial element of $\ker P_n \cong \mu_n^{k_v}/\mu_n$ (Proposition 4.2). Then $(x')^* = x^*$. One checks this by noting $\zeta^* = 1$: every symbol $(\zeta_a, \zeta_b)_n$ is a power of $(\zeta_n, \zeta_n)_n = 1$. The condition is then $J(\zeta x, f_v) = J(x, f_v)$. Please state this case explicitly. It exists only for non-field tori ($k_v \geq 2$).

The unit element satisfies it on split tori. Let t be regular in $T(\mathcal{O}_v)$ for the split torus, with $v \nmid n$, and use the standard measures. Then

$$O(t, 1_{K_v}) = \prod_{\alpha>0} |1-\alpha(t)|^{-1} \qquad \text{and} \qquad \Delta(t) = \prod_{\alpha>0} |1-\alpha(t)|,$$

so the normalized orbital integral is $J(t, 1_{K_v}) = 1$ for all such t. In particular $J(\zeta t) = J(t)$. Hence:

- *For $r = 2$* the only tori are split ($k_v = 2$) and fields ($k_v = 1$, where the condition is vacuous). So the unramified unit elements are FK-compatible at every unramified split place. This makes $r = 2$, $v \nmid n$, a realistic first case in which to prove hypothesis (i) of Proposition 8.3 outright.

- *For $r \geq 3$* there are mixed tori $F \times E'$, and compatibility of $1_{K_v Z_v}$ there is not clear. Translating by a nonscalar root of unity moves an element near the center away from it, and the normalized orbital integral of the unit element is not constant near the center. This should be checked, or extracted from [FK, §§11–12], before (i) is asserted with $\lambda_v = 1$ at almost all places.

1.4 Two small gaps in the proof of Proposition 8.3

(a) *Rational lift versus star.* The proof uses that the covered term at the rational class $\iota(s^n)$ is the local star s_v^* at every v. This holds because the rational splitting ι is a homomorphism: $\iota(s^n) = \iota(s)^n$, and $\iota(s)_v$ is a lift of s_v. Add this sentence.

(b) *Complex places.* Hypothesis (i) is imposed at the complex places, where the cover splits. There, the compatibility and the pointwise identity say that $J(x, f_\infty)$ is a pullback of an orbital-integral function by $x \mapsto x^n$. So the same singular-power issue as in the "regularity and germ obligation" paragraph arises at $x = \mathrm{diag}(1, \zeta, \ldots)$. That paragraph should say it applies at the archimedean places too.

1.5 Lemma 8.2

Lemma 8.2 is correct, and its proof is a one-line consequence of the central transformation law. One hypothesis is redundant: "The corresponding identity holds also at central translates" follows if the region is required to be stable under Z_v, which it should be anyway. With §1.3, the lemma's content is the nonscalar kernel case.

2. Sections 2–7: rechecks

- *Proposition 4.2.* The new proof is correct.
  - $P_n^{-1}(U^n) = K_0 U$ and $\mathrm{vol}(K_0 U) = |K_0| \,\mathrm{vol}(U)$.
  - $[U : U^n] = |n|_F^{-(r-1)}$ from the $\mathbb{Z}_p$-rank $(r-1)[F : \mathbb{Q}_p]$.

  One point deserves a clause: the image of principal units in $E^\times / F^\times$ can contain torsion, e.g. $\zeta_p \in E \setminus F$. "Sufficiently small" handles this, because with compatible levels $U_E^{(em)} \cap F = U_F^{(m)}$ the quotient lattice is saturated, hence torsion-free. Remark 4.3 and its $\mathbb{Z}$ counterexample are now right.

- *§2.2.* The argument $u(x) \in \{\pm 1\} \cap \mu_n = \{1\}$ is correct. The global statement $\theta|_{K^\times} = 1$ is correct.

- *Remark 3.3.* Correct, including the reason $\zeta_3 \notin F^{\times 3}$ (no ζ_9 in $\mathbb{Q}_3(\zeta_3)$), and the field-torus proof.

- *Proposition 7.1.* Correct. One suggestion: the compatibility $\omega_\pi(a) = \omega_{\widetilde{\pi}}(a^*)$ need not be assumed. Replace x by ax in Zou's (3.2), with a central, using $(ax)^* = a^* x^*$ and that the roots of $(ax)^n$ are at. This gives the compatibility whenever $\theta_{\widetilde{\pi}}$ is not identically zero on the elements x^* with x^n regular. One line would make the proposition self-contained.

- *§8.4.* The argument is now internally consistent: no real places, the covers split at complex places, and Clozel–Delorme covers the linear case. See §0 for what I could not verify.

3. Editorial points

(1) *Undefined symbol.* I_F in the display of wild constants in §6.3 (source line 616) is never defined. Write "$[E^\times : F^\times E^{\times 3}] = 9$" or drop it.

(2) *"New".* The title of §4, "A *new* local power-map calculation", overstates. The index formula is classical; what is useful here is the Haar modulus and the bookkeeping. Drop "new".

(3) *Duplicated display.* The display for $c(\gamma)$ appears twice, in Remark 5.3 and §8.1. Keep one.

(4) *Labels.* The status convention now matches Part I. Its claim "as in Parts I–III" depends on Part III, Draft 9, which I have not seen. Part III, Draft 6, used **[O]** for "open". Please confirm.

(5) *Wording.* The acknowledgments call the paper a "research note"; elsewhere it is "Part IV". Otherwise the acknowledgments are fine, and the drafting history is now out of the body.

(6) *Abstract.* It is accurate. It could add a sentence describing Proposition 8.3 as reducing the $D^\times$-versus-cover comparison to explicit split-place conditions.

4. Recommendation

- *Sections 2–7:* accept, after the editorial points.
- *Section 8:* minor revision:
  - restate Proposition 8.3 with the constant Λ (§1.1);
  - simplify its hypotheses using compactness (§1.2);
  - correct the "in particular" of Definition 8.1 and Lemma 8.2 (§1.3);
  - add the two sentences of §1.4.

The most promising next mathematical steps.

(a) *Prove (i) of Proposition 8.3 for $r = 2$ at places $v \nmid n$.* The unit element is FK-compatible there (§1.3), and only two torus types occur. A pointwise identity $J(x^*, \widetilde{f}_v) = \lambda_v J(x, f_v)$ for the unit elements is then a fundamental-lemma statement. I expect [FK, §12] to supply it, since $(n, N_2) = (n, 1) = 1$ holds automatically, but I have not checked that FK's statement there has exactly this pointwise form.

(b) *Compute Λ in that case,* and test it against the identity-term heuristic of §1.2.

(c) *Carry out the finite computation for $(2, 3, \mathbb{Q}_3(\zeta_3))$* (§6.3). It is still the cleanest way to decide the first wild case.

Items (a) and (b) would give the first unconditional instance of the $D^\times$-versus-cover comparison, with test functions restricted at S and at the places dividing n.